

Risk Register

Resilio — Mobile Mental Health Platform for Nepal | Milan Shrestha | ID: 23057135 | 01_Inception/Risk_Register.md

Identified during the Inception and Elaboration phases. Risks were reassessed at the start of each Construction iteration. Likelihood and Impact scored 1 (Low) to 5 (High).

RISK SCORING KEY

Score	Likelihood	Impact
1	Very unlikely	Negligible — no effect on delivery
2	Unlikely	Minor — small rework only
3	Possible	Moderate — delays one iteration
4	Likely	Significant — delays multiple iterations
5	Very likely	Critical — threatens project delivery

Risk Priority = Likelihood × Impact

TECHNICAL RISKS

ID	Risk	Likelihood	Impact	Priority	Mitigation Strategy	Outcome
R-T01	WebRTC peer connection fails on mobile networks (NAT traversal)	4	4	16	Prototype in Elaboration; implement STUN first, add TURN fallback if needed	STUN insufficient on some mobile networks — TURN server deployed; resolved in Iteration 5

ID	Risk	Likelihood	Impact	Priority	Mitigation Strategy	Outcome
R-T02	SuperTokens OTP emails go to spam / not delivered	3	4	12	Test with multiple email providers; configure SPF/DKIM on sending domain	Managed SuperTokens service handled deliverability; no significant issue
R-T03	eSewa SDK version incompatibility with Flutter	3	4	12	Prototype payment flow in Elaboration; pin SDK version in pubspec.yaml	Minor SDK issue resolved; version pinned at tested version
R-T04	Supabase RLS policies too restrictive (blocking legitimate queries)	3	3	9	Test each policy thoroughly in Supabase dashboard before connecting Flutter client	Two policies required adjustment during Iteration 3
R-T05	Flutter BLoC state management complexity causes UI bugs	3	3	9	Use Cubit for simpler screens; full BLoC only where event complexity justifies it	Managed by using Cubit for most screens; no significant issues
R-T06	Cloudinary upload fails on slow connections	2	3	6	Add retry logic and upload progress indicator	Implemented retry; no user-reported failures

ID	Risk	Likelihood	Impact	Priority	Mitigation Strategy	Outcome
R-T07	Offline-to-online sync creates data conflicts	3	3	9	Timestamp-based conflict resolution; offline entries flagged with `synced_from_offline`	Implemented timestamp strategy; tested with TC-U029
R-T08	FCM push notifications not delivered on all Android versions	3	2	6	Test on Android 8 through 14; handle notification permission request (Android 13+)	Runtime permission request added for Android 13+
R-T09	WebRTC audio/video quality poor on 3G	3	3	9	Implement adaptive bitrate; document minimum recommended connection	Adaptive quality implemented; 3G sessions functional but lower quality
R-T10	Code generation (Freezed/json_serializable) causes build failures	2	3	6	Pin build_runner and freezed versions; always run `flutter clean`	Occasional build conflicts; resolved with flag

SCHEDULE RISKS

ID	Risk	Likelihood	Impact	Priority	Mitigation Strategy	Outcome
R-S01	WebRTC integration takes longer than one iteration	4	3	12	Allocate Iteration 5 to WebRTC; overflow into Transition if needed	Overflowed by approximately one week into Transition; no critical impact
R-S02	Client availability for UAT is limited	3	3	9	Schedule UAT sessions weeks in advance; provide async feedback option	Both clients participated on schedule
R-S03	Pre-survey takes longer to reach 100+ responses than expected	2	3	6	Distribute across multiple channels simultaneously from day one	166 responses in 2 weeks; no delay
R-S04	Construction iterations run sequentially — any delay compounds	3	4	12	Parallelise admin + therapist dashboard development in Iteration 5	Parallelisation executed; saved approximately one week
R-S05	Report writing underestimated alongside code development	3	3	9	Allocate dedicated report days each week; write chapters concurrently with development	Managed with weekly writing targets

DOMAIN / CLINICAL RISKS

ID	Risk	Likelihood	Impact	Priority	Mitigation Strategy	Outcome
R-D01	PHQ-9 or GAD-7 scoring implemented incorrectly	2	5	10	Cross-reference Kroenke (2001) and Spitzer (2006) scoring tables; Dr. Singh review	Dr. Singh confirmed correct implementation in UAT
R-D02	Crisis threshold set at wrong PHQ-9 score	2	5	10	Clinically validate threshold with Dr. Singh; use published severe range (≥ 20)	Confirmed correct at ≥ 20
R-D03	Nepal Mental Health Helpline number is incorrect or outdated	2	4	8	Verify 1660-01-11111 number against current government sources	Verified as current
R-D04	Therapist verification process is insufficient — unqualified person passes	2	5	10	Require document uploads; admin manual review required for approval	Admin review workflow implemented

LEGAL / COMPLIANCE RISKS

ID	Risk	Likelihood	Impact	Priority	Mitigation Strategy	Outcome
R-L01	Storing health data outside Nepal violates emerging data residency requirements	2	3	6	Disclose international storage in privacy notice; monitor legislative developments	Disclosed in privacy notice; no current legal prohibition
R-L02	eSewa payment integration violates financial regulations	1	5	5	Use official eSewa SDK; no card data stored in Resilio	Official SDK used throughout
R-L03	SuperToken's session tokens stored insecurely on device	2	4	8	Use SuperToken's recommended secure storage; do not write tokens to plain storage	SuperToken's SDK manages secure storage natively

RISK REGISTER SUMMARY

Priority Level	Count	Resolved	Outstanding
Critical (20–25)	0	—	—
High (12–19)	5	5	0
Medium (6–11)	13	13	0
Low (1–5)	3	3	0

All identified risks were resolved or mitigated before final submission. No risks materialised into project-threatening incidents. The highest-impact technical risk (R-T01 — WebRTC NAT traversal) required a TURN server deployment that was not in the original plan but was resolved within the available schedule buffer.